

Kartikkeya Bomb

901-281-6664 | bomka-27@rhodes.edu | linkedin.com/in/kartibomb | github.com/kartibomb

USCF Chess Peak Elo: 1731

Education

Rhodes College

Memphis, TN

B.S. in Computer Science, Mathematics, and Economics; GPA: 3.81/4.0

Aug. 2023 - May 2027

- **Coursework:** Data Structures & Algorithms, Operating Systems, Linear Algebra, Econometrics, Differential Equations, and Multivariable Calculus.
- **Activities:** CS 141 Teaching Assistant; Computer Science, Engineering, and Chess clubs; Curb Institute of Music performer.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL

AI Engineering and ML: OpenAI API, PyTorch, Hugging Face, LangChain, LangGraph, OpenCV, ONNX, LangSmith, Claude Code

Frameworks: React, Next.js, Spring Boot, FastAPI, Flask

Data and Infrastructure: PostgreSQL, Redis, Prisma, Kafka, Apache Spark, Docker, Kubernetes, AWS, GCP, Git

Experience

AutoZone - Software Engineering Intern

May 2025 - Aug. 2026 - Memphis, TN

- Built an internal React and TypeScript platform for AI-driven store onboarding, using Spring Boot APIs to validate bulk imports and review proposed operational changes.
- Engineered a Python AI agent that reconciled HR, store, and Bringg data, diagnosed inconsistencies, and executed remediation workflows through structured tool calls.
- Built agent state and recovery logic so AI workflows could verify completed actions, retry failed steps, and avoid repeating successful operations.
- Automated pin-pad reassignment across 7,000 stores with React, C++, and WebSockets, **saving 840 hours and \$133K**.

RK Heart and Vascular Care - Software Engineering Intern

Aug. 2024 - Present - Memphis, TN

- Integrated PracticeSuite with custom AI revenue-cycle platform to automate coding, claim submission, denial management, and payment posting across **300+ monthly claims**.
- Built REST/FHIR workflows with claim-state orchestration and audit logging, reducing manual billing work by **84%**.
- Automated **250+ monthly ride requests** with Python/Flask and Baptist Hospital APIs, cutting booking time by **65%** and failed or duplicate requests by **40%**.

IMC Logistics - Software Engineering Intern

Jan. 2025 - May 2025 - Memphis, TN

- Built React dashboards that surfaced AI-detected anomalies across internal store data and third-party delivery configurations.
- Developed an AI-assisted mapping pipeline that interpreted inconsistent shipment schemas and generated normalized delivery mappings across U.S. and European systems.

Projects

SafeZone - AI Disaster Response Platform | React, Mapbox GL, Supabase, Node.js, OpenAI API

- Built a real-time disaster intelligence platform that mapped live hazards, historical events, and community relief centers through Mapbox and Supabase.
- Integrated OpenAI to transform active disaster data into state-aware safety guidance and automatically notify affected users.
- Streamed disaster and relief-center updates through Supabase Realtime, keeping emergency information synchronized across users.

ReefWatch - Coral-Reef Monitoring Network | Python, PyTorch, ONNX, Go, MQTT, Docker

- Deployed quantized audio-vision models to intermittently connected edge nodes with MQTT telemetry, health monitoring, and remote rollouts.
- Cut uplink traffic by **88%** using confidence-gated uploads, compressed telemetry, and store-and-forward synchronization.
- Added versioned model deployments and automatic rollback to recover edge devices from failed or degraded inference releases.

LynxLifts - AI Campus Rideshare Platform | React Native, PyTorch, PostgreSQL/PostGIS, Redis, Kafka, AWS

- Scaled a campus rideshare platform to **200+ verified students and 800+ rides** with real-time driver tracking, messaging, payments, and push notifications.
- Built a PyTorch AI dispatch engine that ranked driver-passenger matches using route overlap, pickup ETA, schedule compatibility, and historical ride behavior, reducing median wait time by **34%**.
- Trained spatiotemporal demand models to predict ride hotspots and served real-time predictions through FastAPI, Redis, and PostGIS with sub-**150 ms** matching latency.

SportsGuessr - Sports Event Location Game | Kotlin, Android SDK, Retrofit, Google Maps API, Room

- Built a GeoGuessr-style Android game that generated location challenges from sports-event, venue, and geographic metadata.
- Ranked map guesses with Haversine-distance scoring and animated geographic feedback through the Google Maps SDK.
- Implemented coroutine-based API ingestion and Room caching for offline sessions and resilient gameplay.

Public Transportation Index | ArcGIS, Google Maps API, Vercel, Airtable

- **Won a hackathon** by building a route-request platform that mapped resident demand and identified underserved transit areas.